Difference-in-difference

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1 Overview

- Treatment cannot be randomized
- Time-series available:
- DID
- Synthetic control + advancements
- Regression discontinuity design (RDD)

Non-random assignment

- We are still in quasi-experimental or observational settings
  - We have non-rando assignment → there are selection effects
- we still want to estimate causal effects
- The problem with regression, matching, weighting, DML etc:
  - They all require conditional independence (unconfoundedness)
  - They require that ALL necessary covariates to capture the selection mechanism are known and measured
  - No selection on observables
  - Unrealistic in many situations

2 The aspect of time

- So far, we have been mostly looking at outcomes after treatment
  - Treated group realizes potential outcomes $Y_i(1)$
  - Untreated group realizes potential outcomes $Y_i(0)$ with $i \neq j$
- Cross-sectional data
  - Observations are collected from many different subjects at a single point in time – here after treatment
- Time-series data / panel data / multi-level data
  - Multiple observations for an individual
    * e.g., single unit across multiple points in time
    * e.g., different units at multiple points in time
    * e.g., nested multi-level data (e.g., students nested within schools, firms nested within countries)
- Example: weekly sales at different stores

	Week 1	Week 2	...	Week t
Store 1	x_{11}	x_{12}	...	x_{1t}
Store 2	x_{21}	x_{22}	...	x_{2t}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
Store s	x_{s1}	x_{s2}	...	x_{st}

3 Recap: panel data – fixed effect model

- $Y_{st} = \mu_s + \tau_t + \beta_1 T_{st} + \epsilon_{st}$
  - μ_s is the store-level fixed effect
    - All unobserved characteristics of the cross-sectional unit (the store) that vary over s but not over t
    - Differences between stores that are stable over time
  - τ_t is the time-period fixed effect
    - Unobserved time varying effects; varies across time t but not across s (stores)
    - Differences over time that are stable across stores
- It is relatively easy to estimate / eliminate μ_s and τ_t if we have sufficient data on both the time and subject dimension (e.g., dummy variables, demeaning, first-differences, etc.)
- β_1 is consistent if $\text{cov}(T_{st}, \epsilon_{st} = 0)$ after controlling for μ_s and τ_t.
- Panel data solves the endogeneity problem that arises from time-invariant or subject-invariant unobserved effects
- Typical subject-specific components
  - Quality of products, location of stores, management capabilities of firms, culture within firms
  - Characteristics of regions, countries, ...
  - Individual customer characteristics
  - Subject fixed-effects are sufficient as long as these characteristics do not vary over time (within the time-frame under investigation)
- Typical time-varying components
  - Seasonality, trend, holidays
  - Events (political, spots), changes in legislation
  - Time fixed-events are sufficient as long as these events affect all cross-sectional units (e.g., all stores) the same way

4 Recap: panel data – remaining endogeneity problems

- Oanel data (or multi-level data more generally) is already a very powerful way of addressing a wide range of endogeneity concerns
- Problems that remain: time and subject specific omitted variables and reverse causality
  - Unobserved characteristics of the cross-section units s that vary over time t and correlate with $T_s \rightarrow$ i.e., X_{st} with $\text{cov}(X_{st}, T_{st}) \neq 0$ and influence Y_{st}
  - The outcome variable in a period (or more typically the expectation of it) influences the treatment assignment T_{st}
    - e.g., local and time dependent events that influence sales and also influence treatment selection as managers expect higher/lower sales
    - e.g., loacl competitor behavior that is store specific and time varying and influences treatment selection
    - e.g., treatment selection based on local trends (increasing or decreasing sales)

5 Solutions (so far)

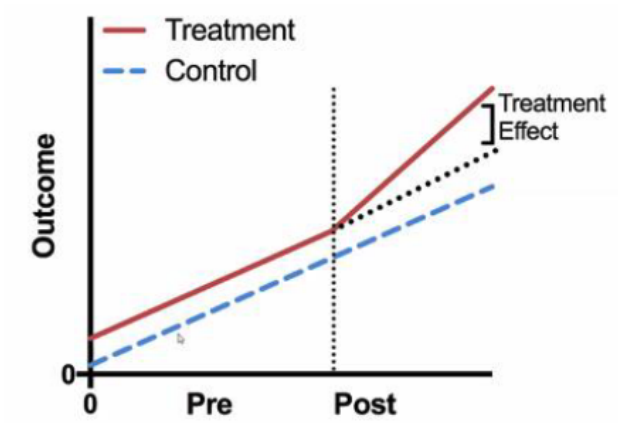
- If we are able to observe these variables, we should add them as controls
  - Selection on observables
  - $Y_{st} = \mu_s + \tau_t + \beta_1 + T_{st} + \gamma X_{st} \epsilon_{st}$
  - We can use regression analysis (typical panel estimators) or matching, weighting, DML
- If we are not able to observe these variables, we may use instrumental variables
  - Selection on unobservables
  - Instrumental variables would also need to vary over s and t
  - We can use 2SLS or the control function approach

6 Difference-in-difference (DiD) framework

- We now focus on two groups with measurements before and after treatment
  - One group is treated, and the other is not (binary treatment)
- Remember, within-subject framework utilizes potential outcomes before and after treatment for the same individual i to calculate treatment effects
  - $\mathbb{E}[Y_{i,\text{post}}] - \mathbb{E}[Y_{i,\text{pre}}]$
  - However, this difference only represents a causal effect if $Y_{i,\text{pre}}(0) = Y_{i,\text{post}}(0)$
    - i.e., the counterfactual outcome does not change over time
  - Typically, we cannot rule out that other factors contributed to pre - post difference
    - Trends, events, policy changes etc.
- We will now extend this idea, relaxing the assumption of no change over time to a more general parallel trends assumption

7 DiD rationale

- Untangle the impact of the treatment from a general trend
- What is the difference in outcomes that we would observe for the treated group (red line) relative to the counterfactual (black dotted line) if the trend that we observed for the control and treated group would continue?
- **Core idea:**
  - There can be level differences between treatment and control (e.g., $\alpha_{g=1} - \alpha_{g=0}$), but the treatment units would have followed the same path parallel to the control units in the absence of treatment (e.g., $\delta_{\text{post}} - \delta_{\text{pre}}$)
  - Common trend or parallel trend assumption



8 Core of DiD framwork

- We assume additive structure for mean potential outcomes in case of no treatment:

$$\mathbb{E}[Y_{igt}(0)] = \delta_t + \alpha_g$$

  - There are effects of time δ_t and treatment group α_g and these are linearly separable
- Remember: we can represent the observed outcomes as a dunction of the potential outcomes:

$$Y_{igt} = Y_{igt}(1)T_{gt} + Y_{igt}(0)(1 - T_{gt})$$

- Taking expectation and rewriting gives:

$$\mathbb{E}[Y_{igt}] = \beta T_{gt} + \delta_t + \alpha_g$$

- We compare outcomes for treated and control *after* intervention
- And compare outcomes for treated before and after intervention
- DiD estimator:

$$\text{ATET}_{\text{DID}} = \beta$$

9 Common or parallel trend assumption

- For the simple 2 groups - 2 time points setting the assumption is formally given by

$$\mathbb{E}[Y_{igt}(0)|t = \text{post}, g = 0] - \mathbb{E}[Y_{igt}(0)|t = \text{pre}, g = 0]$$

  - The difference between pre and post (potential) outcomes is equal between treatment and control group (and under the typical assumption of SUTVA this relates to observed outcomes)

- What happens when the assumption fails?

$$\text{ATET}_{\text{DID}} = \beta + (\text{difference in trends})$$

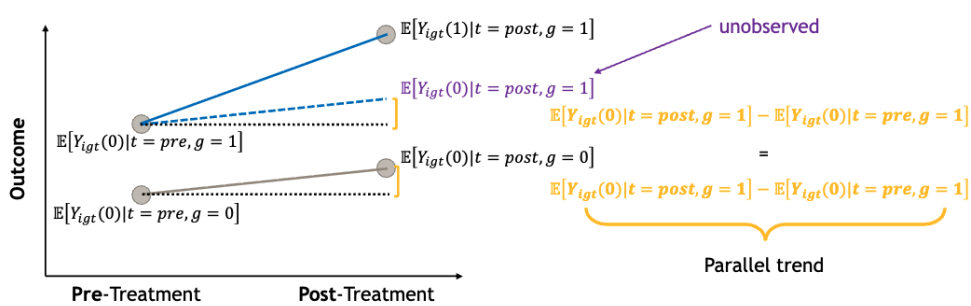
More generally,

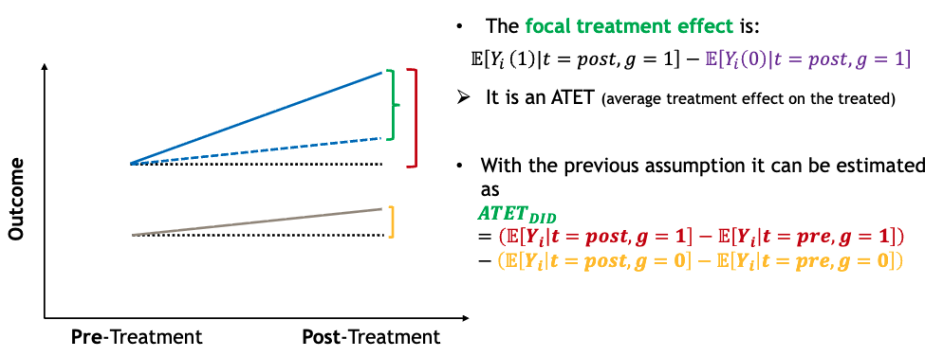
- while traditional independence assumption $\{Y_i(1), Y_i(0)\} \perp T_i$ states that the treatment assignment can't be related to the levels of potential outcomes
  - i.e., $\alpha_{g=1} - \alpha_{g=0} = 0$ because this is $\mathbb{E}[Y_{it}(0)|g = 1] - \mathbb{E}[Y_{it}(0)|g = 0] = 0$
  - i.e., $\delta_{\text{post}} - \delta_{\text{pre}} = 0$ because this is $\mathbb{E}[Y_{it}(0)|\text{post}] - \mathbb{E}[Y_{it}(0)|\text{pre}] = 0$
- The DiD relaxes this assumption and only requires that the treatment assignmetn can't be related to the growth in potential outcomes over time

$$\{Y_{i,t}(0) - Y_{i,t-1}(0)\} \perp T_i$$
$$\{\delta_t - \delta_{t-1}\} \perp T_i$$

Example

- Simple setting: 2 groups - 2 time points





10 Common trend assumption

- How to assess common trends?
  - Graphical inspection of parallel trends in pre-treatment periods (figures like before)
  - Estimating trend models on the pre-treatment periods and assessing the time trend differences between the control and treatment groups (e.g., using an interaction)
    - $Y_{it} = \alpha + \gamma_1 t + \gamma_2 T_i t + \xi_{it} \quad \text{for } t < \text{POST}_t$
  - Some studies use DiD in combination with some form of matching
    - e.g., create groups with matching trends